

GraphTheorySymbol:

A set of $\text{\LaTeX}$ commands to have symbols designed for
Graph Theory

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Abstract

This documentation is for $\text{\LaTeX}$ package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols, and the meta-logical operators symbols were created by the author of this package.

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1 Introduction

This documentation is for $\text{\LaTeX}$ package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols, and the meta-logical operators symbols were created by the author of this package.

This package is licenced under Lesser GNU General Public License version 3 or later (LGPLv3+). It means you are free do anything with it, but if you do improve the code of the symbols that we provide, please do share these improvements in a similar way. There is no restriction on other graph theory symbols/code, even if it would be a good fit for inclusion in this package. There is also no need to share in a similar way what you build with this package. The text of the licenses is available at:

<https://www.gnu.org/licenses/>.

The git repository of this source code is also available at:

<https://github.com/LLyaudet/GraphTheorySymbol/>.

Clone it with this URL:

[git@github.com:LLyaudet/GraphTheorySymbol.git](https://github.com:LLyaudet/GraphTheorySymbol.git).

This package is available on CTAN:

<https://ctan.org/pkg/graph-theory-symbol>.

This is our first $\text{\LaTeX}$ package. It may be that our macros are quite slow and that a document making extensive use of these macros will be slow to compile. If you know how to produce better $\text{\LaTeX}$ code, please submit a pull request on GitHub.

2 Usage

2.1 Memo or for the impatient

Add in your preamble: `\usepackage{graph-theory-symbol}`.

Long name	Short name	Result	In context
<code>\GTSadjacency/</code>	<code>\GTSa/</code>	$\circ$	$u \circ v$
<code>\GTSadjacencyRelation/</code>	<code>\GTSaR/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacency/</code>	<code>\GTSu/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyRelation/</code>	<code>\GTSuR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUp/</code>	<code>\GTSdAU/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpRelation/</code>	<code>\GTSdAUR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUp/</code>	<code>\GTSnDAU/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpRelation/</code>	<code>\GTSnDAUR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDown/</code>	<code>\GTSdAD/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownRelation/</code>	<code>\GTSdADR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDown/</code>	<code>\GTSnDAD/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownRelation/</code>	<code>\GTSnDADR/</code>	$\circ$	$u \circ v$
<code>\GTSincidence/</code>	<code>\GTSi/</code>	$\circ$	$v \circ e$
<code>\GTSincidenceRelation/</code>	<code>\GTSiR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidence/</code>	<code>\GTSnI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceRelation/</code>	<code>\GTSnIR/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidence/</code>	<code>\GTSoI/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceRelation/</code>	<code>\GTSoIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidence/</code>	<code>\GTSnOI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceRelation/</code>	<code>\GTSnOIR/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidence/</code>	<code>\GTSiI/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceRelation/</code>	<code>\GTSiIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidence/</code>	<code>\GTSnII/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceRelation/</code>	<code>\GTSnIIR/</code>	$\circ$	$v \circ e$
Long name	Short name	Result	In context
<code>\GTSadjacencyMonochrome/</code>	<code>\GTSaM/</code>	$\circ$	$u \circ v$
<code>\GTSadjacencyMonochromeRelation/</code>	<code>\GTSaMR/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyMonochrome/</code>	<code>\GTSuM/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyMonochromeRelation/</code>	<code>\GTSuMR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpMonochrome/</code>	<code>\GTSdAUM/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSdAUMR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpMonochrome/</code>	<code>\GTSnDAUM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSnDAUMR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownMonochrome/</code>	<code>\GTSdADM/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSdADMR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownMonochrome/</code>	<code>\GTSnDADM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSnDADMR/</code>	$\circ$	$u \circ v$
<code>\GTSincidenceMonochrome/</code>	<code>\GTSiM/</code>	$\circ$	$v \circ e$
<code>\GTSincidenceMonochromeRelation/</code>	<code>\GTSiMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceMonochrome/</code>	<code>\GTSnIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceMonochromeRelation/</code>	<code>\GTSnIMR/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceMonochrome/</code>	<code>\GTSoIM/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceMonochromeRelation/</code>	<code>\GTSoIMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceMonochrome/</code>	<code>\GTSnOIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceMonochromeRelation/</code>	<code>\GTSnOIMR/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceMonochrome/</code>	<code>\GTSiIM/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceMonochromeRelation/</code>	<code>\GTSiIMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceMonochrome/</code>	<code>\GTSnIIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceMonochromeRelation/</code>	<code>\GTSnIIMR/</code>	$\circ$	$v \circ e$

\GTSsetMainColor{blue}			
Long name	Short name	Result	In context
\GTSadjacency/	\GTSa/		u  v
\GTSadjacencyRelation/	\GTSaR/		u  v
\GTSunadjacency/	\GTSu/		u  v
\GTSunadjacencyRelation/	\GTSuR/		u  v
\GTSdirectedAdjacencyUp/	\GTSdAU/		u  v
\GTSdirectedAdjacencyUpRelation/	\GTSdAUR/		u  v
\GTSnegatedDirectedAdjacencyUp/	\GTSnDAU/		u  v
\GTSnegatedDirectedAdjacencyUpRelation/	\GTSnDAUR/		u  v
\GTSdirectedAdjacencyDown/	\GTSdAD/		u  v
\GTSdirectedAdjacencyDownRelation/	\GTSdADR/		u  v
\GTSnegatedDirectedAdjacencyDown/	\GTSnDAD/		u  v
\GTSnegatedDirectedAdjacencyDownRelation/	\GTSnDADR/		u  v
\GTSincidence/	\GTSi/		v  e
\GTSincidenceRelation/	\GTSiR/		v  e
\GTSnegatedIncidence/	\GTSnI/		v  e
\GTSnegatedIncidenceRelation/	\GTSnIR/		v  e
\GTSoutIncidence/	\GTSoI/		v  e
\GTSoutIncidenceRelation/	\GTSoIR/		v  e
\GTSnegatedOutIncidence/	\GTSnOI/		v  e
\GTSnegatedOutIncidenceRelation/	\GTSnOIR/		v  e
\GTSinIncidence/	\GTSiI/		v  e
\GTSinIncidenceRelation/	\GTSiIR/		v  e
\GTSnegatedInIncidence/	\GTSnII/		v  e
\GTSnegatedInIncidenceRelation/	\GTSnIIR/		v  e

2.2 Detailed usage

First you need to add in your preamble: `\usepackage{graph-theory-symbol}`.

All symbols macros were designed with ease of use in mind: they are called with a trailing slash, they can be called in text mode or in math mode, a variant with relation spacing for math mode exists with suffix “relation”.

`\GTS@newcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

`{\langle commandname \rangle}` The name of the command to create.

`{\langle commandcode \rangle}` The code of the command to create.

`\GTS@newprotectedcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

`{\langle commandname \rangle}` The name of the command to create.

`{\langle commandcode \rangle}` The code of the command to create.

`\GTSsetMainColor` This macro will set the main color used in GraphTheorySymbol. This color has initial value “black”.

Argument:

`{\langle color \rangle}` The color that must be used by GraphTheorySymbol afterward.

`\GTSsetSecondColor` This macro will set the second color used in GraphTheorySymbol. This color has initial value “gray”.

Argument:

`{\langle color \rangle}` The color that must be used by GraphTheorySymbol afterward.

`\GTS@a` This macro will draw any of the adjacency symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and `\GTS@b`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.

Arguments:

`{\langle mainColor \rangle}` If non-empty, this color will be used as main color.

`{\langle secondColor \rangle}` If non-empty, this color will be used as second color.

`{\langle drawSlash \rangle}` Non-empty implies that a negation will be drawn.

`{\langle drawUpArrow \rangle}` Non-empty implies that an up-arrow will be drawn.

`{\langle drawDownArrow \rangle}` Non-empty implies that a down-arrow will be drawn.

`\GTS@b` This macro builds on top of `\GTS@a` to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and `\GTS@a`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.

Arguments:

`{\langle mainColor \rangle}` If non-empty, this color will be used as main color.

`{\langle secondColor \rangle}` If non-empty, this color will be used as second color.

`{\langle drawSlash \rangle}` Non-empty implies that a negation will be drawn.

`{\langle drawUpArrow \rangle}` Non-empty implies that an up-arrow will be drawn.

`{\langle drawDownArrow \rangle}` Non-empty implies that a down-arrow will be drawn.

`{\langle relation \rangle}` Non-empty implies that relation spacing will be added.

`\GTSxa` Versatile macros for adjacency that can be given options.

`\GTSxadj` Argument:

`\GTSxadjacency` [`\langle optionsList \rangle`] Sets the following options if not empty:

mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,

secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,

monochrome or mono or mon or m (equivalent to mC=black,sC=black),

negated or not or n,

up or u,

down or d,

relation or rel or r.

`\GTSadjacency` This macro will draw an adjacency symbol. A first drawing didn’t include the small

`\GTSadj` dots in the middle of the two circles corresponding to the graph vertices. But it was too

`\GTSa` similar to “spoon” and multimap symbols, and could be confused with a graph with only two vertices and one edge. Hence, these dots were added and they have the nice property

to remind of the classical textual notation of adjacency relation as Adj(...).

A.\GTSadjacency/A.
or A.\GTSadj/A.
or A.\GTSa/A.
will yield $A \circ A$.
u \GTSadjacency/ v
or u \GTSadj/ v
or u \GTSa/ v
will yield $u \circ v$.
\$u \GTSadjacency/ v\$
or \$u \GTSadj/ v\$
or \$u \GTSa/ v\$
will yield $u \circ v$.

\GTSadjacencyRelation This macro will draw an adjacency symbol.

\GTSadjRel \$u\GTSadjacencyRelation/v\$
\GTSaR or \$u\GTSadjRel/v\$
or \$u\GTSaR/v\$
will yield $u \circ v$.

\GTSunadjacency This macro will draw an unadjacency symbol.

\GTSunadj A.\GTSunadjacency/A.
\GTSu or A.\GTSunadj/A.
or A.\GTSu/A.
will yield $A \circ A$.
u \GTSunadjacency/ v
or u \GTSunadj/ v
or u \GTSu/ v
will yield $u \circ v$.
\$u \GTSunadjacency/ v\$
or \$u \GTSunadj/ v\$
or \$u \GTSu/ v\$
will yield $u \circ v$.

\GTSunadjacencyRelation This macro will draw an unadjacency symbol.

\GTSunadjRel \$u\GTSunadjacencyRelation/v\$
\GTSuR or \$u\GTSunadjRel/v\$
or \$u\GTSuR/v\$
will yield $u \circ v$.

\GTSdirectedAdjacencyUp This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

A.\GTSdirectedAdjacencyUp/A.
or A.\GTSdirAdjUp/A.
or A.\GTSdAU/A.
will yield $A \circ A$.
u \GTSdirectedAdjacencyUp/ v
or u \GTSdirAdjUp/ v
or u \GTSdAU/ v
will yield $u \circ v$.
\$u \GTSdirectedAdjacencyUp/ v\$
or \$u \GTSdirAdjUp/ v\$
or \$u \GTSdAU/ v\$
will yield $u \circ v$.

\GTSdirectedAdjacencyUp-Relation This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

\GTSdirAdjUpRel \$u\GTSdirectedAdjacencyUpRelation/v\$
\GTSdAUR or \$u\GTSdirAdjUpRel/v\$
or \$u\GTSdAUR/v\$
will yield $u \circ v$.

\GTSnegatedDirected-AdjacencyUp This macro will draw a negated directed adjacency upward symbol.

\GTSnegDirAdjUp
\GTSnDAU

$A.\backslash\text{GTSnegatedDirectedAdjacencyUp}/A.$
 or $A.\backslash\text{GTSnegDirAdjUp}/A.$
 or $A.\backslash\text{GTSnDAU}/A.$
 will yield $A.\overline{\wedge}A.$
 $u\ \backslash\text{GTSnegatedDirectedAdjacencyUp}/\ v$
 or $u\ \backslash\text{GTSnegDirAdjUp}/\ v$
 or $u\ \backslash\text{GTSnDAU}/\ v$
 will yield $u\ \overline{\wedge}\ v.$
 $\$u\ \backslash\text{GTSnegatedDirectedAdjacencyUp}/\ v\$$
 or $\$u\ \backslash\text{GTSnegDirAdjUp}/\ v\$$
 or $\$u\ \backslash\text{GTSnDAU}/\ v\$$
 will yield $u\overline{\wedge}v.$

$\backslash\text{GTSnegatedDirectedAdjacencyUpRelation}$ This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.
 $\backslash\text{GTSnegDirAdjUpRel}$ $\$u\backslash\text{GTSnegatedDirectedAdjacencyUpRelation}/v\$$
 $\backslash\text{GTSnDAUR}$ or $\$u\backslash\text{GTSnegDirAdjUpRel}/v\$$
 or $\$u\backslash\text{GTSnDAUR}/v\$$
 will yield $u\ \overline{\wedge}\ v.$

$\backslash\text{GTSdirectedAdjacencyDown}$ This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.
 $\backslash\text{GTSdirAdjDown}$ $A.\backslash\text{GTSdirectedAdjacencyDown}/A.$
 or $A.\backslash\text{GTSdirAdjDown}/A.$
 or $A.\backslash\text{GTSnDAD}/A.$
 will yield $A.\overline{\wedge}A.$
 $u\ \backslash\text{GTSdirectedAdjacencyDown}/\ v$
 or $u\ \backslash\text{GTSdirAdjDown}/\ v$
 or $u\ \backslash\text{GTSnDAD}/\ v$
 will yield $u\ \overline{\wedge}\ v.$
 $\$u\ \backslash\text{GTSdirectedAdjacencyDown}/\ v\$$
 or $\$u\ \backslash\text{GTSdirAdjDown}/\ v\$$
 or $\$u\ \backslash\text{GTSnDAD}/\ v\$$
 will yield $u\overline{\wedge}v.$

$\backslash\text{GTSdirectedAdjacencyDownRelation}$ This macro will draw a directed adjacency downward symbol with relation spacing in math mode.
 $\backslash\text{GTSdirAdjDownRel}$ $\$u\backslash\text{GTSdirectedAdjacencyDownRelation}/v\$$
 $\backslash\text{GTSdADR}$ or $\$u\backslash\text{GTSdirAdjDownRel}/v\$$
 or $\$u\backslash\text{GTSdADR}/v\$$
 will yield $u\ \overline{\wedge}\ v.$

$\backslash\text{GTSnegatedDirectedAdjacencyDown}$ This macro will draw a negated directed adjacency downward symbol.
 $\backslash\text{GTSnegDirAdjDown}$ $A.\backslash\text{GTSnegatedDirectedAdjacencyDown}/A.$
 or $A.\backslash\text{GTSnegDirAdjDown}/A.$
 $\backslash\text{GTSnDAD}$ or $A.\backslash\text{GTSnDAD}/A.$
 will yield $A.\overline{\wedge}A.$
 $u\ \backslash\text{GTSnegatedDirectedAdjacencyDown}/\ v$
 or $u\ \backslash\text{GTSnegDirAdjDown}/\ v$
 or $u\ \backslash\text{GTSnDAD}/\ v$
 will yield $u\ \overline{\wedge}\ v.$
 $\$u\ \backslash\text{GTSnegatedDirectedAdjacencyDown}/\ v\$$
 or $\$u\ \backslash\text{GTSnegDirAdjDown}/\ v\$$
 or $\$u\ \backslash\text{GTSnDAD}/\ v\$$
 will yield $u\overline{\wedge}v.$

$\backslash\text{GTSnegatedDirectedAdjacencyDownRelation}$ This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.
 $\backslash\text{GTSnegDirAdjDownRel}$ $\$u\backslash\text{GTSnegatedDirectedAdjacencyDownRelation}/v\$$
 $\backslash\text{GTSnDADR}$ or $\$u\backslash\text{GTSnegDirAdjDownRel}/v\$$
 or $\$u\backslash\text{GTSnDADR}/v\$$
 will yield $u\ \overline{\wedge}\ v.$

\GTS@i This macro will draw any of the incidence symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and **\GTS@j**: **\GTSxincidence**, **\GTSxinc**, and **\GTSxi**. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

Arguments:

\{<mainColor> If non-empty, this color will be used as main color.
\{<secondColor> If non-empty, this color will be used as second color.
\{<drawSlash> Non-empty implies that a negation will be drawn.
\{<drawUpArrow> Non-empty implies that an up-arrow will be drawn.
\{<drawDownArrow> Non-empty implies that a down-arrow will be drawn.

\GTS@j This macro builds on top of **\GTS@i** to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and **\GTS@i**: **\GTSxincidence**, **\GTSxinc**, and **\GTSxi**.

Arguments:

\{<mainColor> If non-empty, this color will be used as main color.
\{<secondColor> If non-empty, this color will be used as second color.
\{<drawSlash> Non-empty implies that a negation will be drawn.
\{<drawUpArrow> Non-empty implies that an up-arrow will be drawn.
\{<drawDownArrow> Non-empty implies that a down-arrow will be drawn.
\{<relation> Non-empty implies that relation spacing will be added.

\GTSxi Versatile macros for incidence that can be given options.

\GTSxinc Argument:

\GTSxincidence [*<optionsList>*] Sets the following options if not empty:
mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
in or i,
out or o,
relation or rel or r.

\GTSincidence This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

\GTSi **A.\GTSincidence/A.**
or **A.\GTSinc/A.**
or **A.\GTSi/A.**
will yield $A \downarrow A$.
v \GTSincidence/ e
or **v \GTSinc/ e**
or **v \GTSi/ e**
will yield $v \downarrow e$.
\$v \GTSincidence/ e\$
or **\$v \GTSinc/ e\$**
or **\$v \GTSi/ e\$**
will yield $v \downarrow e$.

\GTSincidenceRelation This macro will draw an incidence symbol with relation spacing in math mode.

\GTSincRel **\$v\GTSincidenceRelation/e\$**
\GTSiR or **\$v\GTSincRel/e\$**
or **\$v\GTSiR/e\$**
will yield $v \downarrow e$.

\GTSnegatedIncidence This macro will draw a negated incidence symbol.

\GTSnegInc **A.\GTSnegatedIncidence/A.**
\GTSnI or **A.\GTSnegInc/A.**
or **A.\GTSnI/A.**
will yield $A \not\downarrow A$.
v \GTSnegatedIncidence/ e
or **v \GTSnegInc/ e**
or **v \GTSnI/ e**
will yield $v \not\downarrow e$.

$\$v \setminus \text{GTSnegatedIncidence} / e \$$
 or $\$v \setminus \text{GTSnegInc} / e \$$
 or $\$v \setminus \text{GTSnI} / e \$$
 will yield $v \not_{\rho} e$.

`\GTSnegatedIncidence-` This macro will draw a negated incidence symbol with relation spacing in math mode.
`Relation` `$v\GTSnegatedIncidenceRelation/e$`
`\GTSnegIncRel` or `$v\GTSnegIncRel/e$`
`\GTSnIR` or `$v\GTSnIR/e$`
will yield $v \not\hookrightarrow e$.

`\GTSoutIncidence` This macro will draw a directed incidence upward (out(ward) incidence) symbol. The
`\GTSoutInc` operand on the left of the symbol should be considered the vertex to which the arc/the
`\GTSOI` operand on the right of the symbol is incident.
`A.\GTSoutIncidence/A.`
or `A.\GTSoutInc/A.`
or `A.\GTSOI/A.`
will yield $A.\overset{\uparrow}{\hookrightarrow}A.$
`v \GTSoutIncidence/ e`
or `v \GTSoutInc/ e`
or `v \GTSOI/ e`
will yield $v \overset{\uparrow}{\hookrightarrow} e.$
`$v \GTSoutIncidence/ e$`
or `$v \GTSoutInc/ e$`
or `$v \GTSOI/ e$`
will yield $v\overset{\uparrow}{\hookrightarrow}e.$

`\GTSoutIncidenceRelation` This macro will draw a directed incidence upward (out(ward) incidence) symbol with
`\GTSoutIncRel` relation spacing in math mode.
`\GTSOIR` `$v\GTSoutIncidenceRelation/e$`
or `$v\GTSoutIncRel/e$`
or `$v\GTSOIR/e$`
will yield $v \overset{\uparrow}{\hookrightarrow} e.$

`\GTSnegatedOutIncidence` This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.
`\GTSnegOutInc` `A.\GTSnegatedOutIncidence/A.`
`\GTSnOI` or `A.\GTSnegOutInc/A.`
or `A.\GTSnOI/A.`
will yield $A.\not\overset{\uparrow}{\hookrightarrow}A.$
`v \GTSnegatedOutIncidence/ e`
or `v \GTSnegOutInc/ e`
or `v \GTSnOI/ e`
will yield $v \not\overset{\uparrow}{\hookrightarrow} e.$
`$v \GTSnegatedOutIncidence/ e$`
or `$v \GTSnegOutInc/ e$`
or `$v \GTSnOI/ e$`
will yield $v\not\overset{\uparrow}{\hookrightarrow}e.$

`\GTSnegatedOutIncidence-` This macro will draw a negated directed incidence upward (out(ward) incidence)
`Relation` symbol with relation spacing in math mode.
`\GTSnegOutIncRel` `$v\GTSnegatedOutIncidenceRelation/e$`
`\GTSnOIR` or `$v\GTSnegOutIncRel/e$`
or `$v\GTSnOIR/e$`
will yield $v \not\overset{\uparrow}{\hookrightarrow} e.$

`\GTSinIncidence` This macro will draw a directed incidence downward (in(ward) incidence) symbol. The
`\GTSinInc` operand on the left of the symbol should be considered the vertex to which the arc/the
`\GTSiI` operand on the right of the symbol is incident.
`A.\GTSinIncidence/A.`
or `A.\GTSinInc/A.`
or `A.\GTSiI/A.`
will yield $A.\overset{\downarrow}{\hookrightarrow}A.$
`v \GTSinIncidence/ e`
or `v \GTSinInc/ e`
or `v \GTSiI/ e`
will yield $v \overset{\downarrow}{\hookrightarrow} e.$
`$v \GTSinIncidence/ e$`
or `$v \GTSinInc/ e$`

or $\$v \backslash \text{GTSiI} / e\$$
will yield $v \overset{\circ}{\circ} e$.

$\backslash \text{GTSinIncidenceRelation}$ This macro will draw a directed incidence downward (in(ward) incidence) symbol with
 $\backslash \text{GTSinIncRel}$ relation spacing in math mode.

$\backslash \text{GTSiIR}$ $\$v \backslash \text{GTSinIncidenceRelation} / e\$$
or $\$v \backslash \text{GTSinIncRel} / e\$$
or $\$v \backslash \text{GTSiIR} / e\$$
will yield $v \overset{\circ}{\circ} e$.

$\backslash \text{GTSnegatedInIncidence}$ This macro will draw a negated directed incidence downward (in(ward) incidence) sym-
 $\backslash \text{GTSnegInInc}$ bol.

$\backslash \text{GTSnII}$ $A. \backslash \text{GTSnegatedInIncidence} / A.$
or $A. \backslash \text{GTSnegInInc} / A.$
or $A. \backslash \text{GTSnII} / A.$
will yield $A. \overset{\circ}{\circ} A.$
 $v \backslash \text{GTSnegatedInIncidence} / e$
or $v \backslash \text{GTSnegInInc} / e$
or $v \backslash \text{GTSnII} / e$
will yield $v \overset{\circ}{\circ} e.$
 $\$v \backslash \text{GTSnegatedInIncidence} / e\$$
or $\$v \backslash \text{GTSnegInInc} / e\$$
or $\$v \backslash \text{GTSnII} / e\$$
will yield $v \overset{\circ}{\circ} e.$

$\backslash \text{GTSnegatedInIncidence-}$ This macro will draw a negated directed incidence downward (in(ward) incidence)
 Relation symbol with relation spacing in math mode.

$\backslash \text{GTSnegInIncRel}$ $\$v \backslash \text{GTSnegatedInIncidenceRelation} / e\$$
 $\backslash \text{GTSnIIR}$ or $\$v \backslash \text{GTSnegInIncRel} / e\$$
or $\$v \backslash \text{GTSnIIR} / e\$$
will yield $v \overset{\circ}{\circ} e.$

$\backslash \text{GTSadjacencyMonochrome}$ This macro will draw a monochrome adjacency symbol.

$\backslash \text{GTSadjMono}$ $A. \backslash \text{GTSadjacencyMonochrome} / A.$
 $\backslash \text{GTSadjMon}$ or $A. \backslash \text{GTSadjMono} / A.$
 $\backslash \text{GTSaM}$ or $A. \backslash \text{GTSadjMon} / A.$
or $A. \backslash \text{GTSaM} / A.$
will yield $A. \overset{\circ}{\circ} A.$
 $u \backslash \text{GTSadjacencyMonochrome} / v$
or $u \backslash \text{GTSadjMono} / v$
or $u \backslash \text{GTSadjMon} / v$
or $u \backslash \text{GTSaM} / v$
will yield $u \overset{\circ}{\circ} v.$
 $\$u \backslash \text{GTSadjacencyMonochrome} / v\$$
or $\$u \backslash \text{GTSadjMono} / v\$$
or $\$u \backslash \text{GTSadjMon} / v\$$
or $\$u \backslash \text{GTSaM} / v\$$
will yield $u \overset{\circ}{\circ} v.$

$\backslash \text{GTSadjacencyMonochrome-}$ This macro will draw a monochrome adjacency symbol with relation spacing in math
 Relation mode.

$\backslash \text{GTSadjMonoRel}$ $\$u \backslash \text{GTSadjacencyMonochromeRelation} / v\$$
 $\backslash \text{GTSadjMonRel}$ or $\$u \backslash \text{GTSadjMonoRel} / v\$$
 $\backslash \text{GTSaMR}$ or $\$u \backslash \text{GTSadjMonRel} / v\$$
or $\$u \backslash \text{GTSaMR} / v\$$
will yield $u \overset{\circ}{\circ} v.$

$\backslash \text{GTSunadjacencyMonochrome}$ This macro will draw a monochrome unadjacency symbol.

$\backslash \text{GTSunadjMono}$ $A. \backslash \text{GTSunadjacencyMonochrome} / A.$
 $\backslash \text{GTSunadjMon}$ or $A. \backslash \text{GTSunadjMono} / A.$
 $\backslash \text{GTSuM}$ or $A. \backslash \text{GTSunadjMon} / A.$
or $A. \backslash \text{GTSuM} / A.$
will yield $A. \overset{\circ}{\circ} A.$
 $u \backslash \text{GTSunadjacencyMonochrome} / v$

or u \GTSunadjMono/ v
 or u \GTSunadjMon/ v
 or u \GTSuM/ v
 will yield $u \overset{\circ}{\circ} v$.
 \$u \GTSunadjacencyMonochrome/ v\$
 or \$u \GTSunadjMono/ v\$
 or \$u \GTSunadjMon/ v\$
 or \$u \GTSuM/ v\$
 will yield $u \overset{\circ}{\circ} v$.

\GTSunadjacencyMonochrome- This macro will draw a monochrome unadjacency symbol with relation spacing in
 Relation math mode.

\GTSunadjMonoRel \$u\GTSunadjacencyMonochromeRelation/v\$
 \GTSunadjMonRel or \$u\GTSunadjMonoRel/v\$
 \GTSuMR or \$u\GTSunadjMonRel/v\$
 or \$u\GTSuMR/v\$
 will yield $u \overset{\circ}{\circ} v$.

\GTSdirectedAdjacencyUp- This macro will draw a monochrome directed adjacency upward symbol. The operand
 Monochrome on the left of the symbol should be considered the source of the arc; the operand on the
 \GTSdirAdjUpMono right of the symbol should be considered the target of the arc.

\GTSdirAdjUpMon A.\GTSdirectedAdjacencyUpMonochrome/A.
 \GTSdAUM or A.\GTSdirAdjUpMono/A.
 or A.\GTSdirAdjUpMon/A.
 or A.\GTSdAUM/A.
 will yield $A \overset{\circ}{\circ} A$.
 u \GTSdirectedAdjacencyUpMonochrome/ v
 or u \GTSdirAdjUpMono/ v
 or u \GTSdirAdjUpMon/ v
 or u \GTSdAUM/ v
 will yield $u \overset{\circ}{\circ} v$.
 \$u \GTSdirectedAdjacencyUpMonochrome/ v\$
 or \$u \GTSdirAdjUpMono/ v\$
 or \$u \GTSdirAdjUpMon/ v\$
 or \$u \GTSdAUM/ v\$
 will yield $u \overset{\circ}{\circ} v$.

\GTSdirectedAdjacencyUp- This macro will draw a monochrome directed adjacency upward symbol with relation
 MonochromeRelation spacing in math mode.

\GTSdirAdjUpMonoRel \$u\GTSdirectedAdjacencyUpMonochromeRelation/v\$
 \GTSdirAdjUpMonRel or \$u\GTSdirAdjUpMonoRel/v\$
 \GTSdAUMR or \$u\GTSdirAdjUpMonRel/v\$
 or \$u\GTSdAUMR/v\$
 will yield $u \overset{\circ}{\circ} v$.

\GTSnegatedDirected- This macro will draw a monochrome negated directed adjacency upward symbol.

AdjacencyUpMonochrome A.\GTSnegatedDirectedAdjacencyUpMonochrome/A.
 \GTSnegDirAdjUpMono or A.\GTSnegDirAdjUpMono/A.
 \GTSnegDirAdjUpMon or A.\GTSnegDirAdjUpMon/A.
 \GTSnDAUM or A.\GTSnDAUM/A.
 will yield $A \overset{\circ}{\circ} A$.
 u \GTSnegatedDirectedAdjacencyUpMonochrome/ v
 or u \GTSnegDirAdjUpMono/ v
 or u \GTSnegDirAdjUpMon/ v
 or u \GTSnDAUM/ v
 will yield $u \overset{\circ}{\circ} v$.
 \$u \GTSnegatedDirectedAdjacencyUpMonochrome/ v\$
 or \$u \GTSnegDirAdjUpMono/ v\$
 or \$u \GTSnegDirAdjUpMon/ v\$
 or \$u \GTSnDAUM/ v\$
 will yield $u \overset{\circ}{\circ} v$.

`\GTSnegatedDirected-` This macro will draw a monochrome negated directed adjacency upward symbol with
`AdjacencyUpMonochrome-` relation spacing in math mode.
`Relation` `\GTSnegatedDirectedAdjacencyUpMonochromeRelation/v$`
`\GTSnegDirAdjUpMonoRel` or `\GTSnegDirAdjUpMonoRel/v$`
`\GTSnegDirAdjUpMonRel` or `\GTSnegDirAdjUpMonRel/v$`
`\GTSnDAUMR` or `\GTSnDAUMR/v$`
will yield $u \overset{\circ}{\frown} v$.

`\GTSdirectedAdjacencyDown-` This macro will draw a monochrome directed adjacency downward symbol. The
`Monochrome` operand on the left of the symbol should be considered the target of the arc; the operand
`\GTSdirAdjDownMono` on the right of the symbol should be considered the source of the arc.
`\GTSdirAdjDownMon` `A.\GTSdirectedAdjacencyDownMonochrome/A.`
`\GTSdADM` or `A.\GTSdirAdjDownMono/A.`
or `A.\GTSdirAdjDownMon/A.`
or `A.\GTSdADM/A.`
will yield $A \overset{\circ}{\smile} A$.
`u \GTSdirectedAdjacencyDownMonochrome/ v`
or `u \GTSdirAdjDownMono/ v`
or `u \GTSdirAdjDownMon/ v`
or `u \GTSdADM/ v`
will yield $u \overset{\circ}{\smile} v$.
`$u \GTSdirectedAdjacencyDownMonochrome/ v$`
or `$u \GTSdirAdjDownMono/ v$`
or `$u \GTSdirAdjDownMon/ v$`
or `$u \GTSdADM/ v$`
will yield $u \overset{\circ}{\smile} v$.

`\GTSdirectedAdjacencyDown-` This macro will draw a monochrome directed adjacency downward symbol with
`MonochromeRelation` relation spacing in math mode.
`\GTSdirAdjDownMonoRel` `\GTSdirectedAdjacencyDownMonochromeRelation/v$`
`\GTSdirAdjDownMonRel` or `\GTSdirAdjDownMonoRel/v$`
`\GTSdADMR` or `\GTSdirAdjDownMonRel/v$`
or `\GTSdADMR/v$`
will yield $u \overset{\circ}{\smile} v$.

`\GTSnegatedDirected-` This macro will draw a monochrome negated directed adjacency downward symbol.
`AdjacencyDownMonochrome` `A.\GTSnegatedDirectedAdjacencyDownMonochrome/A.`
`\GTSnegDirAdjDownMono` or `A.\GTSnegDirAdjDownMono/A.`
`\GTSnegDirAdjDownMon` or `A.\GTSnegDirAdjDownMon/A.`
`\GTSnDADM` or `A.\GTSnDADM/A.`
will yield $A \overset{\circ}{\frown} A$.
`u \GTSnegatedDirectedAdjacencyDownMonochrome/ v`
or `u \GTSnegDirAdjDownMono/ v`
or `u \GTSnegDirAdjDownMon/ v`
or `u \GTSnDADM/ v`
will yield $u \overset{\circ}{\frown} v$.
`$u \GTSnegatedDirectedAdjacencyDownMonochrome/ v$`
or `$u \GTSnegDirAdjDownMono/ v$`
or `$u \GTSnegDirAdjDownMon/ v$`
or `$u \GTSnDADM/ v$`
will yield $u \overset{\circ}{\frown} v$.

`\GTSnegatedDirected-` This macro will draw a monochrome negated directed adjacency downward symbol
`AdjacencyDownMonochrome-` with relation spacing in math mode.
`Relation` `\GTSnegatedDirectedAdjacencyDownMonochromeRelation/v$`
`\GTSnegDirAdjDownMonoRel` or `\GTSnegDirAdjDownMonoRel/v$`
`\GTSnegDirAdjDownMonRel` or `\GTSnegDirAdjDownMonRel/v$`
`\GTSnDADMR` or `\GTSnDADMR/v$`
will yield $u \overset{\circ}{\frown} v$.

`\GTSincidenceMonochrome` This macro will draw a monochrome incidence symbol. The operand on the left of the
`\GTSincMono` symbol should be considered the vertex to which the edge/the operand on the right of the
`\GTSincMon` symbol is incident.
`\GTSiM` `A.\GTSincidenceMonochrome/A.`

or A.\GTSincMono/A.
 or A.\GTSincMon/A.
 or A.\GTSiM/A.
 will yield $A \downarrow A$.
 $v \backslash \text{GTSincidenceMonochrome} / e$
 or $v \backslash \text{GTSincMono} / e$
 or $v \backslash \text{GTSincMon} / e$
 or $v \backslash \text{GTSiM} / e$
 will yield $v \downarrow e$.
 $\$v \backslash \text{GTSincidenceMonochrome} / e\$$
 or $\$v \backslash \text{GTSincMono} / e\$$
 or $\$v \backslash \text{GTSincMon} / e\$$
 or $\$v \backslash \text{GTSiM} / e\$$
 will yield $v \downarrow e$.

$\backslash \text{GTSincidenceMonochrome-Relation}$ This macro will draw a monochrome incidence symbol with relation spacing in math mode.

$\backslash \text{GTSincMonoRel}$ $\$v \backslash \text{GTSincidenceMonochromeRelation} / e\$$
 $\backslash \text{GTSincMonRel}$ or $\$v \backslash \text{GTSincMonoRel} / e\$$
 $\backslash \text{GTSiMR}$ or $\$v \backslash \text{GTSincMonRel} / e\$$
 or $\$v \backslash \text{GTSiMR} / e\$$
 will yield $v \downarrow e$.

$\backslash \text{GTSnegatedIncidenceMonochrome}$ This macro will draw a monochrome negated incidence symbol.

$\backslash \text{GTSnegatedIncidenceMonochrome}$ A.\GTSnegatedIncidenceMonochrome/A.
 $\backslash \text{GTSnegIncMono}$ or A.\GTSnegIncMono/A.
 $\backslash \text{GTSnegIncMon}$ or A.\GTSnegIncMon/A.
 $\backslash \text{GTSnIM}$ or A.\GTSnIM/A.
 will yield $A \downarrow A$.
 $v \backslash \text{GTSnegatedIncidenceMonochrome} / e$
 or $v \backslash \text{GTSnegIncMono} / e$
 or $v \backslash \text{GTSnegIncMon} / e$
 or $v \backslash \text{GTSnIM} / e$
 will yield $v \downarrow e$.
 $\$v \backslash \text{GTSnegatedIncidenceMonochrome} / e\$$
 or $\$v \backslash \text{GTSnegIncMono} / e\$$
 or $\$v \backslash \text{GTSnegIncMon} / e\$$
 or $\$v \backslash \text{GTSnIM} / e\$$
 will yield $v \downarrow e$.

$\backslash \text{GTSnegatedIncidenceMonochromeRelation}$ This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.

$\backslash \text{GTSnegIncMonoRel}$ $\$v \backslash \text{GTSnegatedIncidenceMonochromeRelation} / e\$$
 $\backslash \text{GTSnegIncMonRel}$ or $\$v \backslash \text{GTSnegIncMonoRel} / e\$$
 $\backslash \text{GTSnIMR}$ or $\$v \backslash \text{GTSnegIncMonRel} / e\$$
 or $\$v \backslash \text{GTSnIMR} / e\$$
 will yield $v \downarrow e$.

$\backslash \text{GTSoutIncidenceMonochrome}$ This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

$\backslash \text{GTSoutIncidenceMonochrome}$ A.\GTSoutIncidenceMonochrome/A.
 or A.\GTSoutIncMono/A.
 or A.\GTSoutIncMon/A.
 or A.\GTSoutIM/A.
 will yield $A \uparrow A$.
 $v \backslash \text{GTSoutIncidenceMonochrome} / e$
 or $v \backslash \text{GTSoutIncMono} / e$
 or $v \backslash \text{GTSoutIncMon} / e$
 or $v \backslash \text{GTSoutIM} / e$
 will yield $v \uparrow e$.
 $\$v \backslash \text{GTSoutIncidenceMonochrome} / e\$$
 or $\$v \backslash \text{GTSoutIncMono} / e\$$

or $\$v \backslash \text{GTSoutIncMon}/ e\$$
or $\$v \backslash \text{GTSOIM}/ e\$$
will yield $v \uparrow e$.

$\backslash \text{GTSoutIncidenceMonochrome-}$ This macro will draw a monochrome directed incidence upward (out(ward) incidence)
Relation symbol with relation spacing in math mode.
 $\backslash \text{GTSoutIncMonoRel}$ $\$v \backslash \text{GTSoutIncidenceMonochromeRelation}/ e\$$
 $\backslash \text{GTSoutIncMonRel}$ or $\$v \backslash \text{GTSoutIncMonoRel}/ e\$$
 $\backslash \text{GTSOIMR}$ or $\$v \backslash \text{GTSoutIncMonRel}/ e\$$
or $\$v \backslash \text{GTSOIMR}/ e\$$
will yield $v \uparrow e$.

$\backslash \text{GTSnegatedOutIncidence-}$ This macro will draw a monochrome negated directed incidence upward (out(ward)
Monochrome incidence) symbol.
 $\backslash \text{GTSnegOutIncMono}$ $A. \backslash \text{GTSnegatedOutIncidenceMonochrome}/ A.$
 $\backslash \text{GTSnegOutIncMon}$ or $A. \backslash \text{GTSnegOutIncMono}/ A.$
 $\backslash \text{GTSnOIM}$ or $A. \backslash \text{GTSnegOutIncMon}/ A.$
or $A. \backslash \text{GTSnOIM}/ A.$
will yield $A. \uparrow A.$
 $v \backslash \text{GTSnegatedOutIncidenceMonochrome}/ e$
or $v \backslash \text{GTSnegOutIncMono}/ e$
or $v \backslash \text{GTSnegOutIncMon}/ e$
or $v \backslash \text{GTSnOIM}/ e$
will yield $v \uparrow e$.
 $\$v \backslash \text{GTSnegatedOutIncidenceMonochrome}/ e\$$
or $\$v \backslash \text{GTSnegOutIncMono}/ e\$$
or $\$v \backslash \text{GTSnegOutIncMon}/ e\$$
or $\$v \backslash \text{GTSnOIM}/ e\$$
will yield $v \uparrow e$.

$\backslash \text{GTSnegatedOutIncidence-}$ This macro will draw a monochrome negated directed incidence upward (out(ward)
MonochromeRelation incidence) symbol with relation spacing in math mode.
 $\backslash \text{GTSnegOutIncMonoRel}$ $\$v \backslash \text{GTSnegatedOutIncidenceMonochromeRelation}/ e\$$
 $\backslash \text{GTSnegOutIncMonRel}$ or $\$v \backslash \text{GTSnegOutIncMonoRel}/ e\$$
 $\backslash \text{GTSnOIMR}$ or $\$v \backslash \text{GTSnegOutIncMonRel}/ e\$$
or $\$v \backslash \text{GTSnOIMR}/ e\$$
will yield $v \uparrow e$.

$\backslash \text{GTSinIncidenceMonochrome}$ This macro will draw a monochrome directed incidence downward (in(ward) incidence)
 $\backslash \text{GTSinIncMono}$ symbol. The operand on the left of the symbol should be considered the vertex to which
 $\backslash \text{GTSinIncMon}$ the arc/the operand on the right of the symbol is incident.
 $\backslash \text{GTSiIM}$ $A. \backslash \text{GTSinIncidenceMonochrome}/ A.$
or $A. \backslash \text{GTSinIncMono}/ A.$
or $A. \backslash \text{GTSinIncMon}/ A.$
or $A. \backslash \text{GTSiIM}/ A.$
will yield $A. \downarrow A.$
 $v \backslash \text{GTSinIncidenceMonochrome}/ e$
or $v \backslash \text{GTSinIncMono}/ e$
or $v \backslash \text{GTSinIncMon}/ e$
or $v \backslash \text{GTSiIM}/ e$
will yield $v \downarrow e$.
 $\$v \backslash \text{GTSinIncidenceMonochrome}/ e\$$
or $\$v \backslash \text{GTSinIncMono}/ e\$$
or $\$v \backslash \text{GTSinIncMon}/ e\$$
or $\$v \backslash \text{GTSiIM}/ e\$$
will yield $v \downarrow e$.

$\backslash \text{GTSinIncidenceMonochrome-}$ This macro will draw a monochrome directed incidence downward (in(ward) incidence)
Relation symbol with relation spacing in math mode.
 $\backslash \text{GTSinIncMonoRel}$ $\$v \backslash \text{GTSinIncidenceMonochromeRelation}/ e\$$
 $\backslash \text{GTSinIncMonRel}$ or $\$v \backslash \text{GTSinIncMonoRel}/ e\$$
 $\backslash \text{GTSiIMR}$ or $\$v \backslash \text{GTSinIncMonRel}/ e\$$
or $\$v \backslash \text{GTSiIMR}/ e\$$
will yield $v \downarrow e$.

`\GTSnegatedInIncidence-` This macro will draw a monochrome negated directed incidence downward (in(ward)
`Monochrome` incidence) symbol.

`\GTSnegInIncMono` `A.\GTSnegatedInIncidenceMonochrome/A.`
`\GTSnegInIncMon` or `A.\GTSnegInIncMono/A.`
`\GTSnIIM` or `A.\GTSnegInIncMon/A.`
or `A.\GTSnIIM/A.`
will yield $A \not\downarrow A$.
`v \GTSnegatedInIncidenceMonochrome/ e`
or `v \GTSnegInIncMono/ e`
or `v \GTSnegInIncMon/ e`
or `v \GTSnIIM/ e`
will yield $v \not\downarrow e$.
`$v \GTSnegatedInIncidenceMonochrome/ e$`
or `$v \GTSnegInIncMono/ e$`
or `$v \GTSnegInIncMon/ e$`
or `$v \GTSnIIM/ e$`
will yield $v \not\downarrow e$.

`\GTSnegatedInIncidence-` This macro will draw a monochrome negated directed incidence downward (in(ward)
`MonochromeRelation` incidence) symbol with relation spacing in math mode.

`\GTSnegInIncMonoRel` `$v\GTSnegatedInIncidenceMonochromeRelation/e$`
`\GTSnegInIncMonRel` or `$v\GTSnegInIncMonoRel/e$`
`\GTSnIIMR` or `$v\GTSnegInIncMonRel/e$`
or `$v\GTSnIIMR/e$`
will yield $v \not\downarrow e$.

`\GTSmetaNot` This macro will draw a meta-not or relation-not symbol.

`\GTSmN` `A.\GTSmetaNot/A.`
or `A.\GTSmN/A.`
will yield $A \not\equiv A$.
`u \GTSmetaNot/\GTSdAU/ v`
or `u \GTSmN/\GTSdAU/ v`
will yield $u \not\equiv v$.
`$u \GTSmetaNot/ \GTSdAU/ v$`
or `$u \GTSmN/ \GTSdAU/ v$`
will yield $u \not\equiv v$.

`\GTSmetaAnd` This macro will draw a meta-and or relations-and symbol. This is your duty to check
`\GTSmA` that all relations have same arity. It should be easy as long as you remain in the realm of
graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup
has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge
symbols have taken various meanings, whilst we want an unambiguous symbol.
Look at the discussion after all logical symbols for less trivial uses and recommendations.

`A.\GTSmetaAnd/A.`
or `A.\GTSmA/A.`
will yield $A \not\wedge A$.
`u \GTSdAD/\GTSmetaAnd/\GTSdAU/ v`
or `u \GTSdAD/\GTSmA/\GTSdAU/ v`
will yield $u \not\wedge v$.
`$u \GTSdAD/ \GTSmetaAnd/ \GTSdAU/ v$`
or `$u \GTSdAD/ \GTSmA/ \GTSdAU/ v$`
will yield $u \not\wedge v$.

`\GTSmetaAndRelation` This macro will draw a meta-and symbol with relation spacing in math mode.

`\GTSmetaAndRel` `$u \GTSdAD/\GTSmetaAndRelation/\GTSdAU/ v$`
`\GTSmAR` or `$u \GTSdAD/\GTSmetaAndRel/\GTSdAU/ v$`
or `$u \GTSdAD/\GTSmAR/\GTSdAU/ v$`
will yield $u \not\wedge v$.

`\GTSmetaOr` This macro will draw a meta-or or relations-or symbol. This is your duty to check that all
`\GTSmO` relations have same arity. It should be easy as long as you remain in the realm of graphs/2-
structures/binary structures. We didn't use Tarski's notation, since square cup has taken
the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols
have taken various meanings, whilst we want an unambiguous symbol.

Look at the discussion after all logical symbols for less trivial uses and recommendations.

`A.\GTSmetaOr/A.`

or `A.\GTSmO/A.`

will yield $A.\bigvee A.$

`u \GTSdAD/\GTSmetaOr/\GTSdAU/ v`

or `u \GTSdAD/\GTSmO/\GTSdAU/ v`

will yield $u \bigvee v.$

`$u \GTSdAD/ \GTSmetaOr/ \GTSdAU/ v$`

or `$u \GTSdAD/ \GTSmO/ \GTSdAU/ v$`

will yield $u \bigvee v.$

`\GTSmetaOrRelation` This macro will draw a meta-or symbol with relation spacing in math mode.

`\GTSmetaOrRel $u \GTSdAD/\GTSmetaOrRelation/\GTSdAU/ v$`

`\GTSmOR` or `$u \GTSdAD/\GTSmetaOrRel/\GTSdAU/ v$`

or `$u \GTSdAD/\GTSmOR/\GTSdAU/ v$`

will yield $u \bigvee v.$

2.3 Good practices

As is customary in algebra and logic, the conjunction/multiplication can be replaced by concatenation, and has higher priority than disjunction. So the following formulas are equivalent; we recommend using the shortest one:

Tournament adjacency:

$\forall u, v, u \bigvee v$

is preferred over

$\forall u, v, u \bigvee \bigwedge v,$

or $\forall u, v, u (\bigvee \bigwedge) \bigvee (\bigwedge \bigvee) v.$

Note the use of macros without Relation suffix above, and adding a single `\mathrel{}` around the composition of all relations:

`\mathrel{\GTSdAD/\GTSnDAU/\GTSmO/\GTSnDAD/\GTSdAU/}.`

This is the spacing practice that we recommend, although for very long compositions, it may be quite unreadable, and then spacing everything with Relation could/should be preferred.

We also recommend using brackets when the composition reflects all of the adjacencies between two vertices:

Tournament adjacency:

$\forall u, v, u [\bigvee] v.$

Tournament adjacency with maybe some colored edge also, who knows?:

$\forall u, v, u [\bigvee] v.$

Directed graph:

$\forall u, v, u [\bigvee \bigwedge \bigvee \bigwedge] v.$

Oriented graph:

$\forall u, v, u [\bigvee \bigwedge \bigvee \bigwedge] v.$

Finally, we recommend prefix notation without “big” symbol for a sequence of consecutive ors or ands (with parentheses if needed):

$\exists u, v, u [\bigvee] v.$

$\exists v, e, v [\bigvee] e.$

3 Implementation

```

1 <*package>
2 \RequirePackage{amsmath}
3 \RequirePackage{amssymb}
4 \RequirePackage{graphicx}
5 \RequirePackage{keyval}
6 \RequirePackage{fdsymbol}
7 \RequirePackage{lua-unicode-math}
8 \RequirePackage{xcolor}
9
10 \def\GTS@mainColor{black}
11 \def\GTS@secondColor{gray}
12 \def\GTS@mainColorBackup{black}
13 \def\GTS@secondColorBackup{gray}

```



```

14 \def\GTS@drawSlash{}
15 \def\GTS@drawUpArrow{}
16 \def\GTS@drawDownArrow{}
17

\GTS@newcommand This macro will enforce that the other macros are called with a trailing slash, avoiding
problems with spaces after macro call.
Arguments:
{\commandname} The name of the command to create.
{\commandcode} The code of the command to create.
18 \@ifdefinable{\GTS@newcommand}{\def\GTS@newcommand#1#2{%
19   \@ifdefinable{#1}{\def#1/{#2}}
20 }}

\GTS@newprotectedcommand This macro will enforce that the other macros are called with a trailing slash, avoiding
problems with spaces after macro call.
Arguments:
{\commandname} The name of the command to create.
{\commandcode} The code of the command to create.
21 \@ifdefinable{\GTS@newprotectedcommand}{\def\GTS@newprotectedcommand#1#2{%
22   \@ifdefinable{#1}{\protected\def#1/{#2}}
23 }}

\GTSsetMainColor This macro will set the main color used in GraphTheorySymbol. This color has initial value
“black”.
Argument:
{\color} The color that must be used by GraphTheorySymbol afterward.
24 \newcommand{\GTSsetMainColor}[1]{%
25   \edef\GTS@mainColorTemp{#1}%
26   \edef\GTS@mainColorBackup{\GTS@mainColor}%
27   \edef\GTS@mainColor{\GTS@mainColorTemp}%
28 }

\GTSsetSecondColor This macro will set the second color used in GraphTheorySymbol. This color has initial
value “gray”.
Argument:
{\color} The color that must be used by GraphTheorySymbol afterward.
29 \newcommand{\GTSsetSecondColor}[1]{%
30   \edef\GTS@secondColorTemp{#1}%
31   \edef\GTS@secondColorBackup{\GTS@secondColor}%
32   \edef\GTS@secondColor{\GTS@secondColorTemp}%
33 }

\GTS@a This macro will draw any adjacency symbol given the correct parameters. Its use is purely
technical. See the macros derived from it and \GTS@b: \GTSxadjacency, \GTSxadj, and
\GTSxa.
Arguments:
{\mainColor} If non-empty, this color will be used as main color.
{\secondColor} If non-empty, this color will be used as second color.
{\drawSlash} Non-empty implies that a negation will be drawn.
{\drawUpArrow} Non-empty implies that an up-arrow will be drawn.
{\drawDownArrow} Non-empty implies that a down-arrow will be drawn.
34 \newcommand{\GTS@a}[5]{%
35   \if#1\empty\else%
36     \GTSsetMainColor{#1}%
37   \fi%
38   \if#2\empty\else%
39     \GTSsetSecondColor{#2}%
40   \fi%
41   \text{\ensuremath{%
42     \raisebox{-0.32ex}{\scalebox{1.5}[1.5]{%
43       %
44       \raisebox{0.7ex}{\scalebox{0.7}[0.7]{%
45         $\mathcolor{\GTS@secondColor}{\circ}$%

```

```

46 }}%
47 \hspace{-0.39ex}%
48 \raisebox{1.05ex}{\scalebox{0.1}[0.1]{%
49 $\mathcolor{\GTS@secondColor}{\bullet}$%
50 }}%
51 \hspace{-0.12ex}%
52 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
53 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
54 }}%
55 \hspace{-0.40ex}%
56 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
57 $\mathcolor{\GTS@secondColor}{\circ}$%
58 }}%
59 \hspace{-0.39ex}%
60 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
61 $\mathcolor{\GTS@secondColor}{\bullet}$%
62 }}%
63 \hspace{0.27ex}%
if GTS@drawUpArrow empty else
64 \if#4\empty\else%
65 \hspace{-0.65ex}%
66 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
67 \textcolor{\GTS@mainColor}{\textasciicircum}%
68 }}%
69 \fi%
if GTS@drawDownArrow empty else
70 \if#5\empty\else%
71 \hspace{-0.65ex}%
72 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
73 \textcolor{\GTS@mainColor}{\textasciicircum}%
74 }}%
75 \fi%
if GTS@drawSlash empty else
76 \if#3\empty\else%
77 \hspace{-0.65ex}%
78 \raisebox{0.25ex}{\scalebox{0.6}[0.6]{%
79 \textcolor{black}{/}%
80 }}%
81 \fi%
82 %
83 }}%
84 }}%
85 \if#1\empty\else%
86 \GTSsetMainColor{\GTS@mainColorBackup}%
87 \fi%
88 \if#2\empty\else%
89 \GTSsetSecondColor{\GTS@secondColorBackup}%
90 \fi%
91 }

```

\GTS@b This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and **\GTS@a**: **\GTSxadjacency**, **\GTSxadj**, and **\GTSxa**.

Arguments:

- {<mainColor>}** If non-empty, this color will be used as main color.
- {<secondColor>}** If non-empty, this color will be used as second color.
- {<drawSlash>}** Non-empty implies that a negation will be drawn.
- {<drawUpArrow>}** Non-empty implies that an up-arrow will be drawn.
- {<drawDownArrow>}** Non-empty implies that a down-arrow will be drawn.
- {<relation>}** Non-empty implies that relation spacing will be added.

```

92 \newcommand{\GTS@b}[6]{%
93 \if#61%
94 \mathrel{\GTS@a{#1}{#2}{#3}{#4}{#5}}%
95 \else%

```

```

96 \GTS@a{#1}{#2}{#3}{#4}{#5}%
97 \fi%
98 }

```

`\GTSxa` Versatile macros for adjacency that can be given options.

`\GTSxadj` Argument:

`\GTSxadjacency` [*optionsList*] Sets the following options if not empty:
 mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
 secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
 monochrome or mono or mon or m (equivalent to mC=black,sC=black),
 negated or not or n,
 up or u,
 down or d,
 relation or rel or r.

```

99 \define@key{GTSxa}{mainColor}{\def\GTS@@mainColor{#1}}
100 \define@key{GTSxa}{mainC}{\def\GTS@@mainColor{#1}}
101 \define@key{GTSxa}{mColor}{\def\GTS@@mainColor{#1}}
102 \define@key{GTSxa}{mC}{\def\GTS@@mainColor{#1}}
103 \define@key{GTSxa}{secondColor}{\def\GTS@@secondColor{#1}}
104 \define@key{GTSxa}{secondC}{\def\GTS@@secondColor{#1}}
105 \define@key{GTSxa}{sColor}{\def\GTS@@secondColor{#1}}
106 \define@key{GTSxa}{sC}{\def\GTS@@secondColor{#1}}
107 \define@key{GTSxa}{monochrome}[1]%
108 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
109 \define@key{GTSxa}{mono}[1]%
110 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
111 \define@key{GTSxa}{mon}[1]%
112 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
113 \define@key{GTSxa}{m}[1]%
114 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
115 \define@key{GTSxa}{negated}[1]{\def\GTS@@drawSlash{1}}
116 \define@key{GTSxa}{not}[1]{\def\GTS@@drawSlash{1}}
117 \define@key{GTSxa}{n}[1]{\def\GTS@@drawSlash{1}}
118 \define@key{GTSxa}{up}[1]{\def\GTS@@drawUpArrow{1}}
119 \define@key{GTSxa}{u}[1]{\def\GTS@@drawUpArrow{1}}
120 \define@key{GTSxa}{down}[1]{\def\GTS@@drawDownArrow{1}}
121 \define@key{GTSxa}{d}[1]{\def\GTS@@drawDownArrow{1}}
122 \define@key{GTSxa}{relation}[1]{\def\GTS@@relation{1}}
123 \define@key{GTSxa}{rel}[1]{\def\GTS@@relation{1}}
124 \define@key{GTSxa}{r}[1]{\def\GTS@@relation{1}}
125
126 \newcommand{\GTSxa}[1] [] {%
127 \def\GTS@@mainColor{}%
128 \def\GTS@@secondColor{}%
129 \def\GTS@@drawSlash{}%
130 \def\GTS@@drawUpArrow{}%
131 \def\GTS@@drawDownArrow{}%
132 \def\GTS@@relation{}%
133 \setkeys{GTSxa}{#1}%
134 \GTS@b{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
135 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
136 }
137
138 \newcommand{\GTSxadj}[1] [] {\GTSxa[#1]}
139
140 \newcommand{\GTSxadjacency}[1] [] {\GTSxa[#1]}

```

`\GTSadjacency` This macro will draw an adjacency symbol.

`\GTSadj` 141 `\GTS@newcommand{\GTSadjacency}{\GTS@a{}{}{}{}}{}`

`\GTSa` 142 `\GTS@newcommand{\GTSadj}{\GTS@a{}{}{}{}}{}`

143 `\GTS@newcommand{\GTSa}{\GTS@a{}{}{}{}}{}`

`\GTSadjacencyRelation` This macro will draw an adjacency symbol with relation spacing in math mode.

`\GTSadjRel` 144 `\GTS@newcommand{\GTSadjacencyRelation}{\mathrel{\GTSa/}}`

`\GTSaR` 145 `\GTS@newcommand{\GTSadjRel}{\mathrel{\GTSa/}}`

146 `\GTS@newcommand{\GTSaR}{\mathrel{\GTSa/}}`

`\GTSunadjacency` This macro will draw an unadjacency symbol.

```

\GTSunadj 147 \GTS@newcommand{\GTSunadjacency}{\GTS@a{}{}{1}{}}
\GTSu     148 \GTS@newcommand{\GTSunadj}{\GTS@a{}{}{1}{}}
          149 \GTS@newcommand{\GTSu}{\GTS@a{}{}{1}{}}

```

`\GTSunadjacencyRelation` This macro will draw an unadjacency symbol with relation spacing in math mode.

```

\GTSunadjRel 150 \GTS@newcommand{\GTSunadjacencyRelation}{\mathrel{\GTSu/}}
\GTSuR       151 \GTS@newcommand{\GTSunadjRel}{\mathrel{\GTSu/}}
          152 \GTS@newcommand{\GTSuR}{\mathrel{\GTSu/}}

```

`\GTSdirectedAdjacencyUp` This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```

          153 \GTS@newcommand{\GTSdirectedAdjacencyUp}{\GTS@a{}{}{1}{}}
          154 \GTS@newcommand{\GTSdirAdjUp}{\GTS@a{}{}{1}{}}
          155 \GTS@newcommand{\GTSdAU}{\GTS@a{}{}{1}{}}

```

`\GTSdirectedAdjacencyUpRelation` This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

```

\GTSdirAdjUpRel 156 \GTS@newcommand{\GTSdirectedAdjacencyUpRelation}{\mathrel{\GTSdAU/}}
\GTSdAUR       157 \GTS@newcommand{\GTSdirAdjUpRel}{\mathrel{\GTSdAU/}}
          158 \GTS@newcommand{\GTSdAUR}{\mathrel{\GTSdAU/}}

```

`\GTSnegatedDirectedAdjacencyUp` This macro will draw a negated directed adjacency upward symbol.

```

          159 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUp}{\GTS@a{}{}{1}{1}{}}
\GTSnegDirAdjUp 160 \GTS@newcommand{\GTSnegDirAdjUp}{\GTS@a{}{}{1}{1}{}}
\GTSnDAU       161 \GTS@newcommand{\GTSnDAU}{\GTS@a{}{}{1}{1}{}}

```

`\GTSnegatedDirectedAdjacencyUpRelation` This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.

```

\GTSnegDirAdjUpRel 162 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpRelation}{%
\GTSnDAUR       163 \mathrel{\GTSnDAU/}%
          164 }
          165 \GTS@newcommand{\GTSnegDirAdjUpRel}{\mathrel{\GTSnDAU/}}
          166 \GTS@newcommand{\GTSnDAUR}{\mathrel{\GTSnDAU/}}

```

`\GTSdirectedAdjacencyDown` This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```

          167 \GTS@newcommand{\GTSdirectedAdjacencyDown}{\GTS@a{}{}{}{1}}
          168 \GTS@newcommand{\GTSdirAdjDown}{\GTS@a{}{}{}{1}}
          169 \GTS@newcommand{\GTSdAD}{\GTS@a{}{}{}{1}}

```

`\GTSdirectedAdjacencyDownRelation` This macro will draw a directed adjacency downward symbol with relation spacing in math mode.

```

\GTSdirAdjDownRel 170 \GTS@newcommand{\GTSdirectedAdjacencyDownRelation}{\mathrel{\GTSdAD/}}
\GTSdADR       171 \GTS@newcommand{\GTSdirAdjDownRel}{\mathrel{\GTSdAD/}}
          172 \GTS@newcommand{\GTSdADR}{\mathrel{\GTSdAD/}}

```

`\GTSnegatedDirectedAdjacencyDown` This macro will draw a negated directed adjacency downward symbol.

```

          173 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDown}{\GTS@a{}{}{1}{1}}
\GTSnegDirAdjDown 174 \GTS@newcommand{\GTSnegDirAdjDown}{\GTS@a{}{}{1}{1}}
\GTSnDAD       175 \GTS@newcommand{\GTSnDAD}{\GTS@a{}{}{1}{1}}

```

`\GTSnegatedDirectedAdjacencyDownRelation` This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.

```

\GTSnegDirAdjDownRel 176 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownRelation}{%
\GTSnDADR       177 \mathrel{\GTSnDAD/}%
          178 }
          179 \GTS@newcommand{\GTSnegDirAdjDownRel}{\mathrel{\GTSnDAD/}}
          180 \GTS@newcommand{\GTSnDADR}{\mathrel{\GTSnDAD/}}

```

`\GTS@i` This macro will draw any incidence symbol given the correct parameters. Its use is purely technical. See the macros derived from it and `\GTS@j`: `\GTSxincidence`, `\GTSxinc`, and `\GTSxi`. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

Arguments:

`{\mainColor}` If non-empty, this color will be used as main color.

`{\secondColor}` If non-empty, this color will be used as second color.

`{\drawSlash}` Non-empty implies that a negation will be drawn.

`{\drawUpArrow}` Non-empty implies that an up-arrow will be drawn.

`{\drawDownArrow}` Non-empty implies that a down-arrow will be drawn.

```

181 \newcommand{\GTS@i}[5]{%
182 \if#1\empty\else%
183 \GTSsetMainColor{#1}%
184 \fi%
185 \if#2\empty\else%
186 \GTSsetSecondColor{#2}%
187 \fi%
188 \text{\ensuremath{%
189 \raisebox{-0.22ex}{\scalebox{2.05}[2.05]{%
190 %
191 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
192 $\mathcolor{\GTS@secondColor}{\circ}$%
193 }}%
194 \hspace{-0.39ex}%
195 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
196 $\mathcolor{\GTS@secondColor}{\bullet}$%
197 }}%
198 \hspace{-0.12ex}%
199 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
200 $\mathcolor{\GTS@secondColor}{\smallblacksquare}$%
201 }}%
202 \hspace{-0.18ex}%
203 \raisebox{0.15ex}{\scalebox{0.2}[0.2]{%
204 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
205 }}%
206 \hspace{0.2ex}%
207 \if#4\empty\else%
208 \hspace{-0.65ex}%
209 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
210 \textcolor{\GTS@secondColor}{\textasciicircum}%
211 }}%
212 \fi%
213 \if#5\empty\else%
214 \hspace{-0.65ex}%
215 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
216 \textcolor{\GTS@secondColor}{\textasciicircum}%
217 }}%
218 \fi%
219 \if#3\empty\else%
220 \hspace{-0.65ex}%
221 \raisebox{0.1ex}{\scalebox{0.5}[0.5]{%
222 \textcolor{black}{/}%
223 }}%
224 \fi%
225 %
226 }}%
227 }}%
228 \if#1\empty\else%
229 \GTSsetMainColor{\GTS@mainColorBackup}%
230 \fi%
```

```

231 \if#2\empty\else%
232 \GTSsetSecondColor{\GTS@secondColorBackup}%
233 \fi%
234 }

```

\GTS@j This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and **\GTS@i**: **\GTSxincidence**, **\GTSxinc**, and **\GTSxi**.

Arguments:

{\mainColor} If non-empty, this color will be used as main color.
{\secondColor} If non-empty, this color will be used as second color.
{\drawSlash} Non-empty implies that a negation will be drawn.
{\drawUpArrow} Non-empty implies that an up-arrow will be drawn.
{\drawDownArrow} Non-empty implies that a down-arrow will be drawn.
{\relation} Non-empty implies that relation spacing will be added.

```

235 \newcommand{\GTS@j}[6]{%
236 \if#61%
237 \mathrel{\GTS@i{#1}{#2}{#3}{#4}{#5}}%
238 \else%
239 \GTS@i{#1}{#2}{#3}{#4}{#5}%
240 \fi%
241 }

```

\GTSxi Versatile macros for incidence that can be given options.

\GTSxinc Argument:

\GTSxincidence [*optionsList*] Sets the following options if not empty:
 mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
 secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
 monochrome or mono or mon or m (equivalent to mC=black,sC=black),
 negated or not or n,
 in or i,
 out or o,
 relation or rel or r.

```

242 \define@key{GTSxi}{mainColor}{\def\GTS@@mainColor{#1}}
243 \define@key{GTSxi}{mainC}{\def\GTS@@mainColor{#1}}
244 \define@key{GTSxi}{mColor}{\def\GTS@@mainColor{#1}}
245 \define@key{GTSxi}{mC}{\def\GTS@@mainColor{#1}}
246 \define@key{GTSxi}{secondColor}{\def\GTS@@secondColor{#1}}
247 \define@key{GTSxi}{secondC}{\def\GTS@@secondColor{#1}}
248 \define@key{GTSxi}{sColor}{\def\GTS@@secondColor{#1}}
249 \define@key{GTSxi}{sC}{\def\GTS@@secondColor{#1}}
250 \define@key{GTSxi}{monochrome}[1]%
251 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
252 \define@key{GTSxi}{mono}[1]%
253 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
254 \define@key{GTSxi}{mon}[1]%
255 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
256 \define@key{GTSxi}{m}[1]%
257 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
258 \define@key{GTSxi}{negated}[1]{\def\GTS@@drawSlash{1}}
259 \define@key{GTSxi}{not}[1]{\def\GTS@@drawSlash{1}}
260 \define@key{GTSxi}{n}[1]{\def\GTS@@drawSlash{1}}
261 \define@key{GTSxi}{out}[1]{\def\GTS@@drawUpArrow{1}}
262 \define@key{GTSxi}{o}[1]{\def\GTS@@drawUpArrow{1}}
263 \define@key{GTSxi}{in}[1]{\def\GTS@@drawDownArrow{1}}
264 \define@key{GTSxi}{i}[1]{\def\GTS@@drawDownArrow{1}}
265 \define@key{GTSxi}{relation}[1]{\def\GTS@@relation{1}}
266 \define@key{GTSxi}{rel}[1]{\def\GTS@@relation{1}}
267 \define@key{GTSxi}{r}[1]{\def\GTS@@relation{1}}
268
269 \newcommand{\GTSxi}[1][]{%
270 \def\GTS@@mainColor{}%
271 \def\GTS@@secondColor{}%
272 \def\GTS@@drawSlash{}%

```

```

273 \def\GTS@@drawUpArrow{%
274 \def\GTS@@drawDownArrow{%
275 \def\GTS@@relation{%
276 \setkeys{GTSxi}{#1}%
277 \GTS@j{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
278 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
279 }
280
281 \newcommand{\GTSxinc}[1][\GTSxi{#1}]
282
283 \newcommand{\GTSxincidence}[1][\GTSxi{#1}]

\GTSincidence This macro will draw an incidence symbol. The operand on the left of the symbol should be
\GTSinc considered the vertex to which the edge/the operand on the right of the symbol is incident.
\GTSi 284 \GTS@newcommand{\GTSincidence}{\GTS@i{}{}{}{}}
285 \GTS@newcommand{\GTSinc}{\GTS@i{}{}{}{}}
286 \GTS@newcommand{\GTSi}{\GTS@i{}{}{}{}}

\GTSincidenceRelation This macro will draw an incidence symbol with relation spacing in math mode.
\GTSincRel 287 \GTS@newcommand{\GTSincidenceRelation}{\mathrel{\GTSi/}}
\GTSiR 288 \GTS@newcommand{\GTSincRel}{\mathrel{\GTSi/}}
289 \GTS@newcommand{\GTSiR}{\mathrel{\GTSi/}}

\GTSnegatedIncidence This macro will draw a negated incidence symbol.
\GTSnegInc 290 \GTS@newcommand{\GTSnegatedIncidence}{\GTS@i{}{}{1}{}{}}
\GTSnI 291 \GTS@newcommand{\GTSnegInc}{\GTS@i{}{}{1}{}{}}
292 \GTS@newcommand{\GTSnI}{\GTS@i{}{}{1}{}{}}

\GTSnegatedIncidence- This macro will draw a negated incidence symbol with relation spacing in math mode.
Relation 293 \GTS@newcommand{\GTSnegatedIncidenceRelation}{\mathrel{\GTSnI/}}
\GTSnegIncRel 294 \GTS@newcommand{\GTSnegIncRel}{\mathrel{\GTSnI/}}
\GTSnIR 295 \GTS@newcommand{\GTSnIR}{\mathrel{\GTSnI/}}

\GTSoutIncidence This macro will draw a directed incidence upward (out(ward) incidence) symbol. The
\GTSoutInc operand on the left of the symbol should be considered the vertex to which the arc/the
\GTSOI operand on the right of the symbol is incident.
296 \GTS@newcommand{\GTSoutIncidence}{\GTS@i{}{}{1}{}{}}
297 \GTS@newcommand{\GTSoutInc}{\GTS@i{}{}{1}{}{}}
298 \GTS@newcommand{\GTSOI}{\GTS@i{}{}{1}{}{}}

\GTSoutIncidenceRelation This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation
\GTSoutIncRel spacing in math mode.
\GTSOIR 299 \GTS@newcommand{\GTSoutIncidenceRelation}{\mathrel{\GTSOI/}}
300 \GTS@newcommand{\GTSoutIncRel}{\mathrel{\GTSOI/}}
301 \GTS@newcommand{\GTSOIR}{\mathrel{\GTSOI/}}

\GTSnegatedOutIncidence This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.
\GTSnegOutInc 302 \GTS@newcommand{\GTSnegatedOutIncidence}{\GTS@i{}{}{1}{1}{}{}}
\GTSnOI 303 \GTS@newcommand{\GTSnegOutInc}{\GTS@i{}{}{1}{1}{}{}}
304 \GTS@newcommand{\GTSnOI}{\GTS@i{}{}{1}{1}{}{}}

\GTSnegatedOutIncidence- This macro will draw a negated directed incidence upward (out(ward) incidence) symbol
Relation with relation spacing in math mode.
\GTSnegOutIncRel 305 \GTS@newcommand{\GTSnegatedOutIncidenceRelation}{\mathrel{\GTSnOI/}}
\GTSnOIR 306 \GTS@newcommand{\GTSnegOutIncRel}{\mathrel{\GTSnOI/}}
307 \GTS@newcommand{\GTSnOIR}{\mathrel{\GTSnOI/}}

\GTSinIncidence This macro will draw a directed incidence downward (in(ward) incidence) symbol. The
\GTSinInc operand on the left of the symbol should be considered the vertex to which the arc/the
\GTSiI operand on the right of the symbol is incident.
308 \GTS@newcommand{\GTSinIncidence}{\GTS@i{}{}{}{1}}
309 \GTS@newcommand{\GTSinInc}{\GTS@i{}{}{}{1}}
310 \GTS@newcommand{\GTSiI}{\GTS@i{}{}{}{1}}

```

`\GTSinIncidenceRelation` This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

`\GTSinIncRel`

```

\GTSiIR 311 \GTS@newcommand{\GTSinIncidenceRelation}{\mathrel{\GTSiI/}}
312 \GTS@newcommand{\GTSinIncRel}{\mathrel{\GTSiI/}}
313 \GTS@newcommand{\GTSiIR}{\mathrel{\GTSiI/}}

```

`\GTSnegatedInIncidence` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.

`\GTSnegInInc`

```

\GTSnII 314 \GTS@newcommand{\GTSnegatedInIncidence}{\GTS@i{}{}{1}{}{1}}
315 \GTS@newcommand{\GTSnegInInc}{\GTS@i{}{}{1}{}{1}}
316 \GTS@newcommand{\GTSnII}{\GTS@i{}{}{1}{}{1}}

```

`\GTSnegatedInIncidence-Relation` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

`\GTSnegInIncRel`

```

\GTSnIIR 317 \GTS@newcommand{\GTSnegatedInIncidenceRelation}{\mathrel{\GTSnII/}}
318 \GTS@newcommand{\GTSnegInIncRel}{\mathrel{\GTSnII/}}
319 \GTS@newcommand{\GTSnIIR}{\mathrel{\GTSnII/}}

```

`\GTSadjacencyMonochrome` This macro will draw a monochrome adjacency symbol.

`\GTSadjMono`

```

\GTSadjMon 320 \GTS@newcommand{\GTSadjacencyMonochrome}{\GTS@a{black}{black}{}{}{}}
\GTSadjMon 321 \GTS@newcommand{\GTSadjMono}{\GTS@a{black}{black}{}{}{}}
\GTSaM 322 \GTS@newcommand{\GTSadjMon}{\GTS@a{black}{black}{}{}{}}
323 \GTS@newcommand{\GTSaM}{\GTS@a{black}{black}{}{}{}}

```

`\GTSadjacencyMonochrome-Relation` This macro will draw a monochrome adjacency symbol with relation spacing in math mode.

`\GTSadjMonoRel`

```

\GTSadjMonRel 324 \GTS@newcommand{\GTSadjacencyMonochromeRelation}{\mathrel{\GTSaM/}}
325 \GTS@newcommand{\GTSadjMonoRel}{\mathrel{\GTSaM/}}
\GTSadjMonRel 326 \GTS@newcommand{\GTSadjMonRel}{\mathrel{\GTSaM/}}
\GTSaMR 327 \GTS@newcommand{\GTSaMR}{\mathrel{\GTSaM/}}

```

`\GTSunadjacencyMonochrome` This macro will draw a monochrome unadjacency symbol.

`\GTSunadjMono`

```

\GTSunadjMon 328 \GTS@newcommand{\GTSunadjacencyMonochrome}{\GTS@a{black}{black}{1}{}{}}
\GTSunadjMon 329 \GTS@newcommand{\GTSunadjMono}{\GTS@a{black}{black}{1}{}{}}
\GTSuM 330 \GTS@newcommand{\GTSunadjMon}{\GTS@a{black}{black}{1}{}{}}
331 \GTS@newcommand{\GTSuM}{\GTS@a{black}{black}{1}{}{}}

```

`\GTSunadjacencyMonochrome-Relation` This macro will draw a monochrome unadjacency symbol with relation spacing in math mode.

`\GTSunadjMonoRel`

```

\GTSunadjMonRel 332 \GTS@newcommand{\GTSunadjacencyMonochromeRelation}{\mathrel{\GTSuM/}}
\GTSunadjMonRel 333 \GTS@newcommand{\GTSunadjMonoRel}{\mathrel{\GTSuM/}}
\GTSuMR 334 \GTS@newcommand{\GTSunadjMonRel}{\mathrel{\GTSuM/}}
335 \GTS@newcommand{\GTSuMR}{\mathrel{\GTSuM/}}

```

`\GTSdirectedAdjacencyUp-Monochrome` This macro will draw a monochrome directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

`\GTSdirAdjUpMono`

```

\GTSdirAdjUpMon 336 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochrome}{%
\GTSdAUM 337 \GTS@a{black}{black}{}{1}{}%
338 }
339 \GTS@newcommand{\GTSdirAdjUpMono}{\GTS@a{black}{black}{}{1}{}{}}
340 \GTS@newcommand{\GTSdirAdjUpMon}{\GTS@a{black}{black}{}{1}{}{}}
341 \GTS@newcommand{\GTSdAUM}{\GTS@a{black}{black}{}{1}{}{}}

```

`\GTSdirectedAdjacencyUp-MonochromeRelation` This macro will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

`\GTSdirAdjUpMonoRel`

```

\GTSdirAdjUpMonRel 342 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochromeRelation}{%
\GTSdAUMR 343 \mathrel{\GTSdAUM/}%
344 }
345 \GTS@newcommand{\GTSdirAdjUpMonoRel}{\mathrel{\GTSdAUM/}}
346 \GTS@newcommand{\GTSdirAdjUpMonRel}{\mathrel{\GTSdAUM/}}
347 \GTS@newcommand{\GTSdAUMR}{\mathrel{\GTSdAUM/}}

```


$\backslash$ GTSnegatedDirected- AdjacencyUpMonochrome- $\backslash$ GTSnegDirAdjUpMono $\backslash$ GTSnegDirAdjUpMon $\backslash$ GTSnDAUM $\backslash$ GTSnegatedDirected- AdjacencyUpMonochrome- Relation $\backslash$ GTSnegDirAdjUpMonRel $\backslash$ GTSnegDirAdjUpMonoRel $\backslash$ GTSnDAUMR $\backslash$ GTSdirectedAdjacencyDown- Monochrome $\backslash$ GTSdirAdjDownMono $\backslash$ GTSdirAdjDownMon $\backslash$ GTSdADM $\backslash$ GTSdirectedAdjacencyDown- MonochromeRelation $\backslash$ GTSdirAdjDownMonoRel $\backslash$ GTSdirAdjDownMonRel $\backslash$ GTSdADMR $\backslash$ GTSnegatedDirected- AdjacencyDownMonochrome $\backslash$ GTSnegDirAdjDownMono $\backslash$ GTSnegDirAdjDownMon $\backslash$ GTSnDADM $\backslash$ GTSnegatedDirected- AdjacencyDownMonochrome- Relation $\backslash$ GTSnegDirAdjDownMonoRel $\backslash$ GTSnegDirAdjDownMonRel $\backslash$ GTSnDADMR $\backslash$ GTSincidenceMonochrome $\backslash$ GTSincMono $\backslash$ GTSincMon $\backslash$ GTSiM $\backslash$ GTSincidenceMonochrome- Relation $\backslash$ GTSincMonoRel $\backslash$ GTSincMonRel $\backslash$ GTSiMR	This macro will draw a monochrome negated directed adjacency upward symbol. 348 $\backslash$GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochrome}{% 349 $\backslash$GTS@a{black}{black}{1}{1}{}} 350 } 351 $\backslash$GTS@newcommand{\GTSnegDirAdjUpMono}{\GTS@a{black}{black}{1}{1}{}} 352 $\backslash$GTS@newcommand{\GTSnegDirAdjUpMon}{\GTS@a{black}{black}{1}{1}{}} 353 $\backslash$GTS@newcommand{\GTSnDAUM}{\GTS@a{black}{black}{1}{1}{}} This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode. 354 $\backslash$GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochromeRelation}{% 355 $\mathrel{\backslash$GTSnDAUM/}}% 356 } 357 $\backslash$GTS@newcommand{\GTSnegDirAdjUpMonoRel}{\mathrel{\backslashGTSnDAUM/}} 358 $\backslash$GTS@newcommand{\GTSnegDirAdjUpMonRel}{\mathrel{\backslashGTSnDAUM/}} 359 $\backslash$GTS@newcommand{\GTSnDAUMR}{\mathrel{\backslashGTSnDAUM/}} This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc. 360 $\backslash$GTS@newcommand{\GTSdirectedAdjacencyDownMonochrome}{% 361 $\backslash$GTS@a{black}{black}{-}{-}{1}}% 362 } 363 $\backslash$GTS@newcommand{\GTSdirAdjDownMono}{\GTS@a{black}{black}{-}{-}{1}} 364 $\backslash$GTS@newcommand{\GTSdirAdjDownMon}{\GTS@a{black}{black}{-}{-}{1}} 365 $\backslash$GTS@newcommand{\GTSdADM}{\GTS@a{black}{black}{-}{-}{1}} This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode. 366 $\backslash$GTS@newcommand{\GTSdirectedAdjacencyDownMonochromeRelation}{% 367 $\mathrel{\backslash$GTSdADM/}}% 368 } 369 $\backslash$GTS@newcommand{\GTSdirAdjDownMonoRel}{\mathrel{\backslashGTSdADM/}} 370 $\backslash$GTS@newcommand{\GTSdirAdjDownMonRel}{\mathrel{\backslashGTSdADM/}} 371 $\backslash$GTS@newcommand{\GTSdADMR}{\mathrel{\backslashGTSdADM/}} This macro will draw a monochrome negated directed adjacency downward symbol. 372 $\backslash$GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochrome}{% 373 $\backslash$GTS@a{black}{black}{1}{-}{1}}% 374 } 375 $\backslash$GTS@newcommand{\GTSnegDirAdjDownMono}{\GTS@a{black}{black}{1}{-}{1}} 376 $\backslash$GTS@newcommand{\GTSnegDirAdjDownMon}{\GTS@a{black}{black}{1}{-}{1}} 377 $\backslash$GTS@newcommand{\GTSnDADM}{\GTS@a{black}{black}{1}{-}{1}} This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode. 378 $\backslash$GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochromeRelation}{% 379 $\mathrel{\backslash$GTSnDADM/}}% 380 } 381 $\backslash$GTS@newcommand{\GTSnegDirAdjDownMonoRel}{\mathrel{\backslashGTSnDADM/}} 382 $\backslash$GTS@newcommand{\GTSnegDirAdjDownMonRel}{\mathrel{\backslashGTSnDADM/}} 383 $\backslash$GTS@newcommand{\GTSnDADMR}{\mathrel{\backslashGTSnDADM/}} This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident. 384 $\backslash$GTS@newcommand{\GTSincidenceMonochrome}{\GTS@i{black}{black}{-}{-}{}} 385 $\backslash$GTS@newcommand{\GTSincMono}{\GTS@i{black}{black}{-}{-}{}} 386 $\backslash$GTS@newcommand{\GTSincMon}{\GTS@i{black}{black}{-}{-}{}} 387 $\backslash$GTS@newcommand{\GTSiM}{\GTS@i{black}{black}{-}{-}{}} This macro will draw a monochrome incidence symbol with relation spacing in math mode. 388 $\backslash$GTS@newcommand{\GTSincidenceMonochromeRelation}{\mathrel{\backslashGTSiM/}} 389 $\backslash$GTS@newcommand{\GTSincMonoRel}{\mathrel{\backslashGTSiM/}} 390 $\backslash$GTS@newcommand{\GTSincMonRel}{\mathrel{\backslashGTSiM/}} 391 $\backslash$GTS@newcommand{\GTSiMR}{\mathrel{\backslashGTSiM/}}
--	---

`\GTSnegatedIncidence-` This macro will draw a monochrome negated incidence symbol.

Monochrome 392 `\GTS@newcommand{\GTSnegatedIncidenceMonochrome}{%`
`\GTSnegIncMono` 393 `\GTS@i{black}{black}{1}{-}{-}%`
`\GTSnegIncMon` 394 `}`
`\GTSnIM` 395 `\GTS@newcommand{\GTSnegIncMono}{\GTS@i{black}{black}{1}{-}{-}}`
396 `\GTS@newcommand{\GTSnegIncMon}{\GTS@i{black}{black}{1}{-}{-}}`
397 `\GTS@newcommand{\GTSnIM}{\GTS@i{black}{black}{1}{-}{-}}`

`\GTSnegatedIncidence-` This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.
MonochromeRelation
`\GTSnegIncMonoRel` 398 `\GTS@newcommand{\GTSnegatedIncidenceMonochromeRelation}{\mathrel{\GTSnIM/}}`
`\GTSnegIncMonRel` 399 `\GTS@newcommand{\GTSnegIncMonoRel}{\mathrel{\GTSnIM/}}`
`\GTSnIMR` 400 `\GTS@newcommand{\GTSnegIncMonRel}{\mathrel{\GTSnIM/}}`
401 `\GTS@newcommand{\GTSnIMR}{\mathrel{\GTSnIM/}}`

`\GTSoutIncidenceMonochrome` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the
`\GTSoutIncMono` arc/the operand on the right of the symbol is incident.
`\GTSoutIncMon`
`\GTSoIM` 402 `\GTS@newcommand{\GTSoutIncidenceMonochrome}{\GTS@i{black}{black}{-}{1}{-}}`
403 `\GTS@newcommand{\GTSoutIncMono}{\GTS@i{black}{black}{-}{1}{-}}`
404 `\GTS@newcommand{\GTSoutIncMon}{\GTS@i{black}{black}{-}{1}{-}}`
405 `\GTS@newcommand{\GTSoIM}{\GTS@i{black}{black}{-}{1}{-}}`

`\GTSoutIncidenceMonochrome-` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.
Relation
`\GTSoutIncMonoRel` 406 `\GTS@newcommand{\GTSoutIncidenceMonochromeRelation}{\mathrel{\GTSoIM/}}`
`\GTSoutIncMonRel` 407 `\GTS@newcommand{\GTSoutIncMonoRel}{\mathrel{\GTSoIM/}}`
`\GTSoIMR` 408 `\GTS@newcommand{\GTSoutIncMonRel}{\mathrel{\GTSoIM/}}`
409 `\GTS@newcommand{\GTSoIMR}{\mathrel{\GTSoIM/}}`

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.
Monochrome
`\GTSnegOutIncMono` 410 `\GTS@newcommand{\GTSnegatedOutIncidenceMonochrome}{%`
`\GTSnegOutIncMon` 411 `\GTS@i{black}{black}{1}{1}{-}%`
`\GTSnOIM` 412 `}`
413 `\GTS@newcommand{\GTSnegOutIncMono}{\GTS@i{black}{black}{1}{1}{-}}`
414 `\GTS@newcommand{\GTSnegOutIncMon}{\GTS@i{black}{black}{1}{1}{-}}`
415 `\GTS@newcommand{\GTSnOIM}{\GTS@i{black}{black}{1}{1}{-}}`

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.
MonochromeRelation
`\GTSnegOutIncMonoRel` 416 `\GTS@newcommand{\GTSnegatedOutIncidenceMonochromeRelation}{%`
`\GTSnegOutIncMonRel` 417 `\mathrel{\GTSnOIM/}%`
`\GTSnOIMR` 418 `}`
419 `\GTS@newcommand{\GTSnegOutIncMonoRel}{\mathrel{\GTSnOIM/}}`
420 `\GTS@newcommand{\GTSnegOutIncMonRel}{\mathrel{\GTSnOIM/}}`
421 `\GTS@newcommand{\GTSnOIMR}{\mathrel{\GTSnOIM/}}`

`\GTSinIncidenceMonochrome` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the
`\GTSinIncMono` arc/the operand on the right of the symbol is incident.
`\GTSinIncMon`
`\GTSiIM` 422 `\GTS@newcommand{\GTSinIncidenceMonochrome}{\GTS@i{black}{black}{-}{-}{1}}`
423 `\GTS@newcommand{\GTSinIncMono}{\GTS@i{black}{black}{-}{-}{1}}`
424 `\GTS@newcommand{\GTSinIncMon}{\GTS@i{black}{black}{-}{-}{1}}`
425 `\GTS@newcommand{\GTSiIM}{\GTS@i{black}{black}{-}{-}{1}}`

`\GTSinIncidenceMonochrome-` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.
Relation
`\GTSinIncMonoRel` 426 `\GTS@newcommand{\GTSinIncidenceMonochromeRelation}{\mathrel{\GTSiIM/}}`
`\GTSinIncMonRel` 427 `\GTS@newcommand{\GTSinIncMonoRel}{\mathrel{\GTSiIM/}}`
`\GTSiIMR` 428 `\GTS@newcommand{\GTSinIncMonRel}{\mathrel{\GTSiIM/}}`
429 `\GTS@newcommand{\GTSiIMR}{\mathrel{\GTSiIM/}}`

`\GTSnegatedInIncidence-` This macro will draw a monochrome negated directed incidence downward (in(ward) inci-
Monochrome dence) symbol.

```

\GTSnegInIncMono 430 \GTS@newcommand{\GTSnegatedInIncidenceMonochrome}{%
\GTSnegInIncMon 431 \GTS@i{black}{black}{1}{1}%
\GTSnIIM 432 }
433 \GTS@newcommand{\GTSnegInIncMono}{\GTS@i{black}{black}{1}{1}}
434 \GTS@newcommand{\GTSnegInIncMon}{\GTS@i{black}{black}{1}{1}}
435 \GTS@newcommand{\GTSnIIM}{\GTS@i{black}{black}{1}{1}}

```

`\GTSnegatedInIncidence-` This macro will draw a monochrome negated directed incidence downward (in(ward) inci-
MonochromeRelation dence) symbol with relation spacing in math mode.

```

\GTSnegInIncMonoRel 436 \GTS@newcommand{\GTSnegatedInIncidenceMonochromeRelation}{%
\GTSnegInIncMonRel 437 \mathrel{\GTSnIIM}/}%
\GTSnIIMR 438 }
439 \GTS@newcommand{\GTSnegInIncMonoRel}{\mathrel{\GTSnIIM}/}
440 \GTS@newcommand{\GTSnegInIncMonRel}{\mathrel{\GTSnIIM}/}
441 \GTS@newcommand{\GTSnIIMR}{\mathrel{\GTSnIIM}/}

```

`\GTSmetaNot` This macro will draw a meta-not or relation-not symbol.

```

\GTSmN 442 \GTS@newprotectedcommand{\GTSmN}{%
443 \text{\ensuremath{%
444 \raisebox{-0.44ex}{\scalebox{1}[1]{%
445 \raisebox{0ex}{\scalebox{1}[1]{\not$}}}%
446 \hspace{-1.46ex}%
447 \raisebox{1.2ex}{\scalebox{0.7}[0.7]{m}}}%
448 \hspace{0.2ex}%
449 }}}%
450 }}%
451 }
452 \GTS@newcommand{\GTSmetaNot}{\GTSmN/}

```

`\GTSmetaAnd` This macro will draw a meta-and or relations-and symbol. This is your duty to check that all
\GTSmA relations have same arity. It should be easy as long as you remain in the realm of graphs/2-
structures/binary structures. We didn't use Tarski's notation, since square cup has taken
the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols
have taken various meanings, whilst we want an unambiguous symbol.

```

453 \GTS@newprotectedcommand{\GTSmA}{%
454 \text{\ensuremath{%
455 \raisebox{-0.44ex}{\scalebox{0.95}[0.95]{%
456 \raisebox{0ex}{\scalebox{1}[1]{\wedge$}}}%
457 \hspace{-1.46ex}%
458 \raisebox{1.5ex}{\scalebox{0.7}[0.7]{m}}}%
459 }}}%
460 }}%
461 }
462 \GTS@newcommand{\GTSmetaAnd}{\GTSmA/}

```

`\GTSmetaAndRelation` This macro will draw a meta-and symbol with relation spacing in math mode.

```

\GTSmetaAndRel 463 \GTS@newcommand{\GTSmetaAndRelation}{\mathrel{\GTSmA/}}
\GTSmAR 464 \GTS@newcommand{\GTSmetaAndRel}{\mathrel{\GTSmA/}}
465 \GTS@newcommand{\GTSmAR}{\mathrel{\GTSmA/}}

```

`\GTSmetaOr` This macro will draw a meta-or or relations-or symbol. This is your duty to check that all
\GTSmO relations have same arity. It should be easy as long as you remain in the realm of graphs/2-
structures/binary structures ;). We didn't use Tarski's notation, since square cup has taken
the preemptive meaning of disjoint union. Similarly, doublevee or other vee symbols have
taken various meanings, whilst we want an unambiguous symbol.

```

466 \GTS@newprotectedcommand{\GTSmO}{%
467 \text{\ensuremath{%
468 \raisebox{-0.44ex}{\scalebox{0.95}[0.95]{%
469 \raisebox{0ex}{\scalebox{1}[1]{\vee$}}}%
470 \hspace{-1.46ex}%
471 \raisebox{1.5ex}{\scalebox{0.7}[0.7]{m}}}%
472 }}}%

```

```

473 }%
474 }
475 \GTS@newcommand{\GTSmetaOr}{\GTSmO/}

\GTSmetaOrRelation This macro will draw a meta-or symbol with relation spacing in math mode.
\GTSmetaOrRel 476 \GTS@newcommand{\GTSmetaOrRelation}{\mathrel{\GTSmO/}}
\GTSmOR 477 \GTS@newcommand{\GTSmetaOrRel}{\mathrel{\GTSmO/}}
478 \GTS@newcommand{\GTSmOR}{\mathrel{\GTSmO/}}

479 \endinput
480 \end{package}

```

4 Acknowledgments

Thank You God! Thank You Father! Thank You Jesus! Thank You Holy-Spirit!

5 Support

This package is gratis and free. But if you can star it on GitHub, it is a nice encouragement.

6 Change History

v0.1.0 – 2026-08-01	same set of symbols	1
General: Initial version	1	
v1.0.0 – 2026-08-26	v1.2.1 – 2026-09-08	
General: First public release	General: smaller tests output	1
v1.1.0 – 2026-08-29	v1.3.0 – 2026-09-08	
General: Added a memo	General: new macros for monochrome	
v1.1.1 – 2026-08-30	symbols	1
General: Small corrections and		
improvements	v1.4.0 – 2026-09-12	1
v1.2.0 – 2026-09-06	General: macros for symbols for logical	
General: CTAN URL, new macros for	operations on relations	1

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